Numerical Methods

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Time Commitment

Lecture	22 hrs	Saturday/Monday
Practicals	18 hrs	Group
Total	**40 hrs**	

Lecturer

General Lecture	Dr. Thai Minh Quan • 081.555.9669 • Thai-minh.quan@usth.edu.vn
Practical work (computer lab)	Dr. Giang Anh Tuan
	Dr. Phan Thanh Hien
	Dr. Pham Duc Binh
	PhD. Tran Anh Tu
	PhD. Nguyen Dinh Vu

Evaluation

Attendance/Attitude	10%
Exercises	10%
Mid-term test	30%
Final exam	50%
Total	**100%**

Content

No	Contents	Hours			Ref./Resources
		Lect.	Exr.	Prc.	
1	**Introduction**	3		1	[1]: Chap 1-2
2	**Roots of Non-linear equations**	3		1.5	[1] Chap 6
3	**Systems of linear equations**	3		2	[1] Chap 9
4	**LU decomposition**	2		2	[1] Chap 10
5	**Curve fitting**	1		1	
6	**Linear Programming**	1		1	[1] Chap 15
7	**Numerical Differentiation and Integration**	2		3	[1] Chap 21-23

References

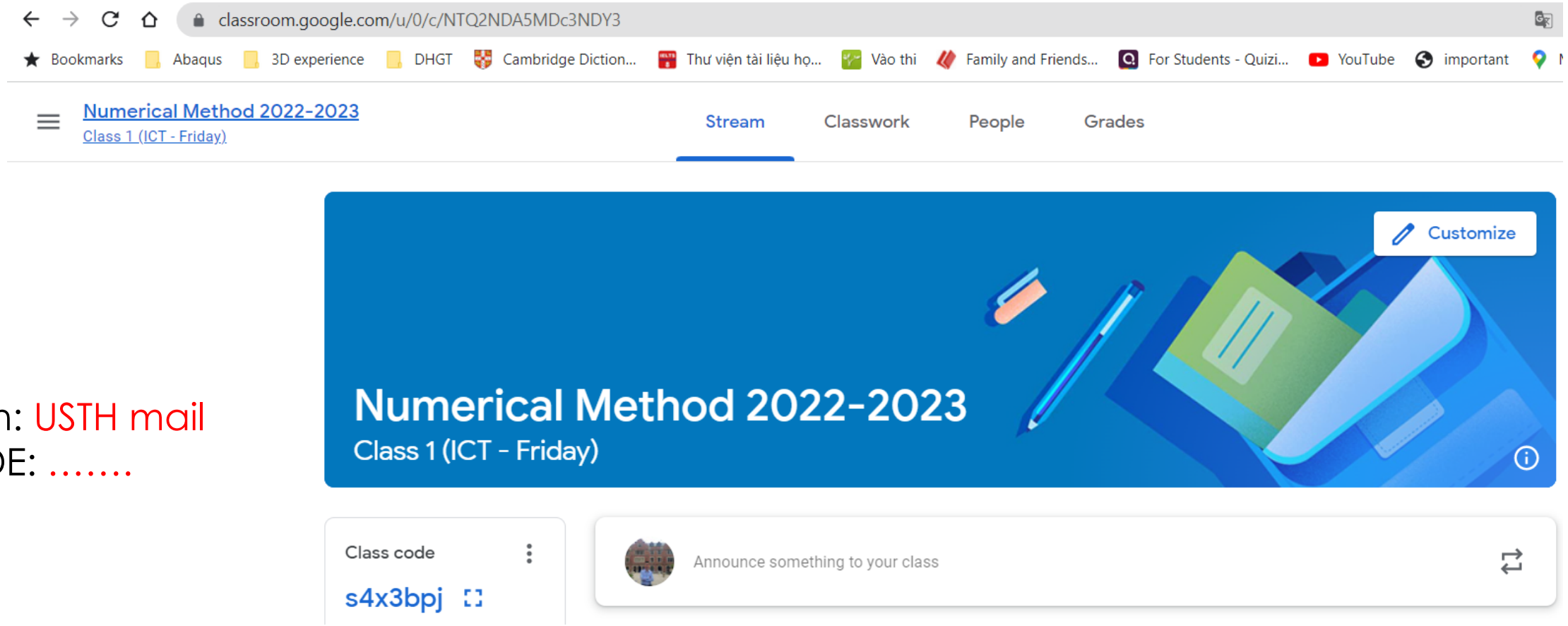
Book: S.C Chapra et al., "Numerical methods for Engineers", 6th Edition, MacGraw-Hill, 2006

Book: Introduction to Numerical Methods and Matlab Programming for Engineers. - *Todd Young and Martin J. Mohlenkamp*

Lecture notes – *Thai Minh Quan in Classroom*

https://classroom.google.com

References



Login: USTH mail
CODE: .....

Motivation

Analytical method

$$x-3=0$$

$$x^2-2x-3=0$$

$$x^3-2x^2-2x+3=0$$

$$x^4+x^3-2x^2-3x+3=0$$

$$x^5-x^4+x^3-2x^2-3x+3=0$$

$$a_n x^n + a_{n-1}x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0 = 0$$











Numerical methods

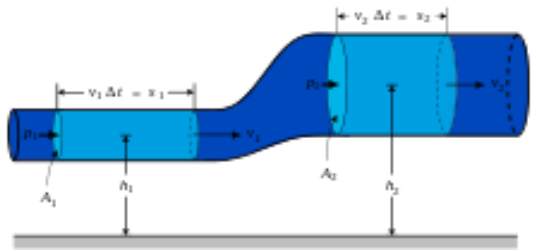


Motivation

Or…

Navier-Stokes momentum equation (*convective form*)

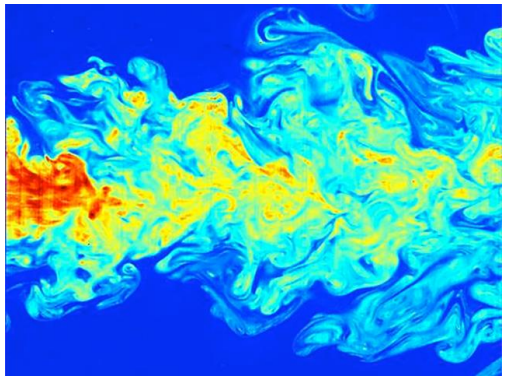
$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho}\nabla \bar{p} + \nu \nabla^2 \mathbf{u} + \frac{1}{3}\nu \nabla (\nabla \cdot \mathbf{u}) + \mathbf{g}.$$

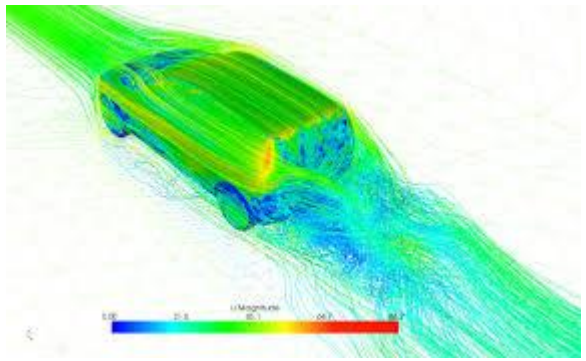


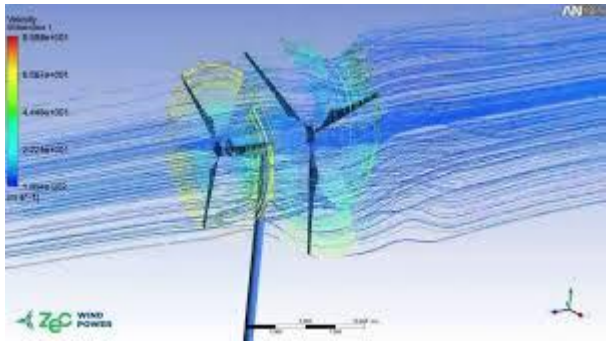
fluid flow

Analytical methods ?

Numerical methods





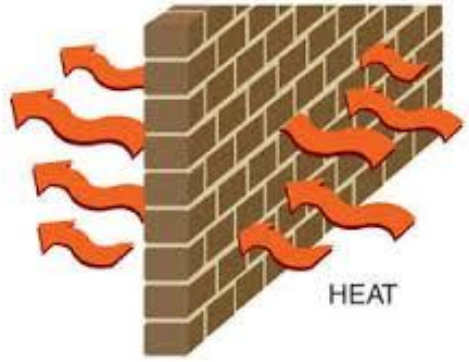


Motivation

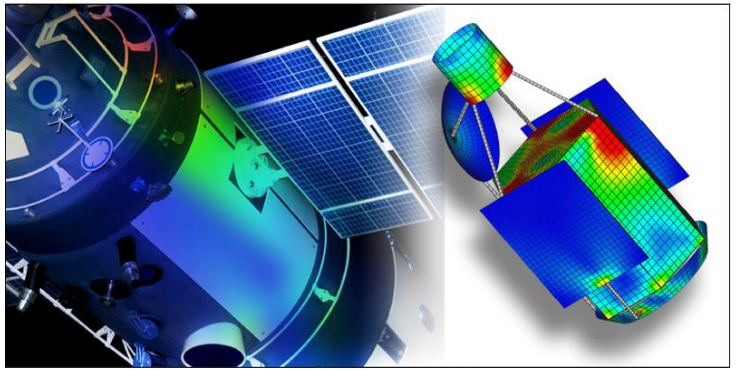
Or…

Fourier's law

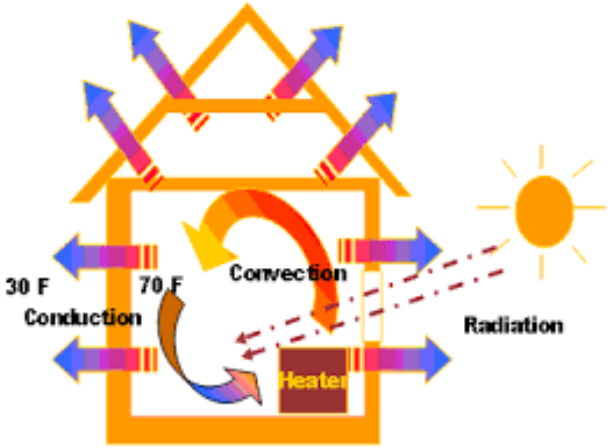
$$\vec{q} = -k\nabla T$$



Analytical methods ?



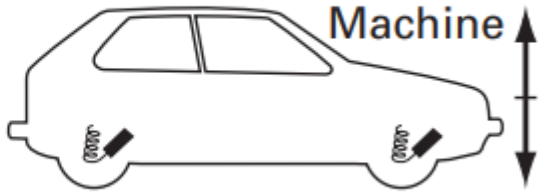
Numerical methods



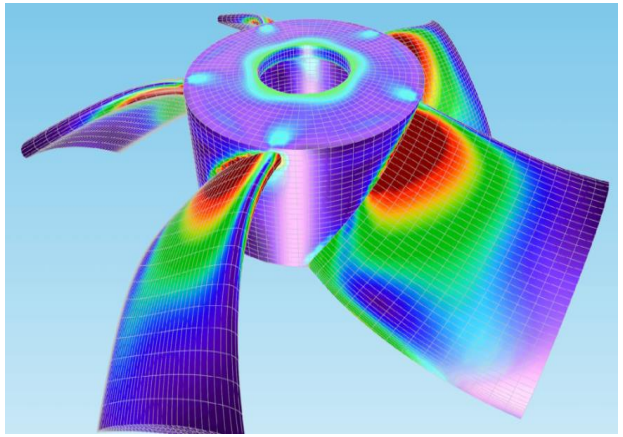
Motivation

Or…

Mechanical engineering

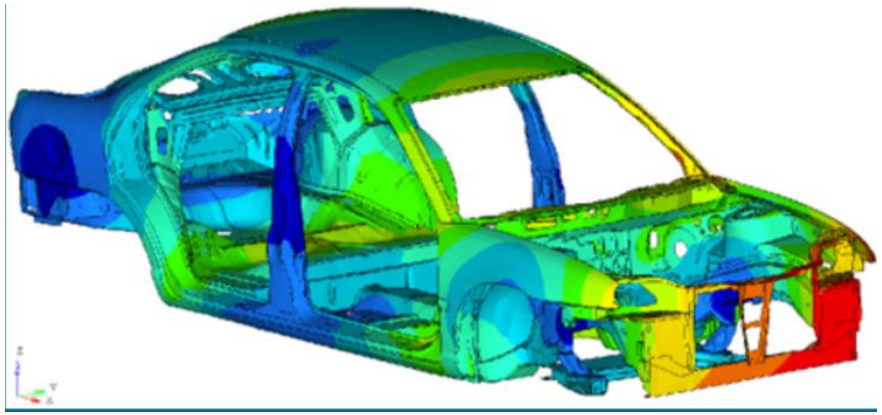


Conservation of momentum



Analytical methods ?

Numerical methods



Reasons to study

- Solve problems with no analytic solution

  Non-linear equations

  Complex behaviors

- Understand these methods

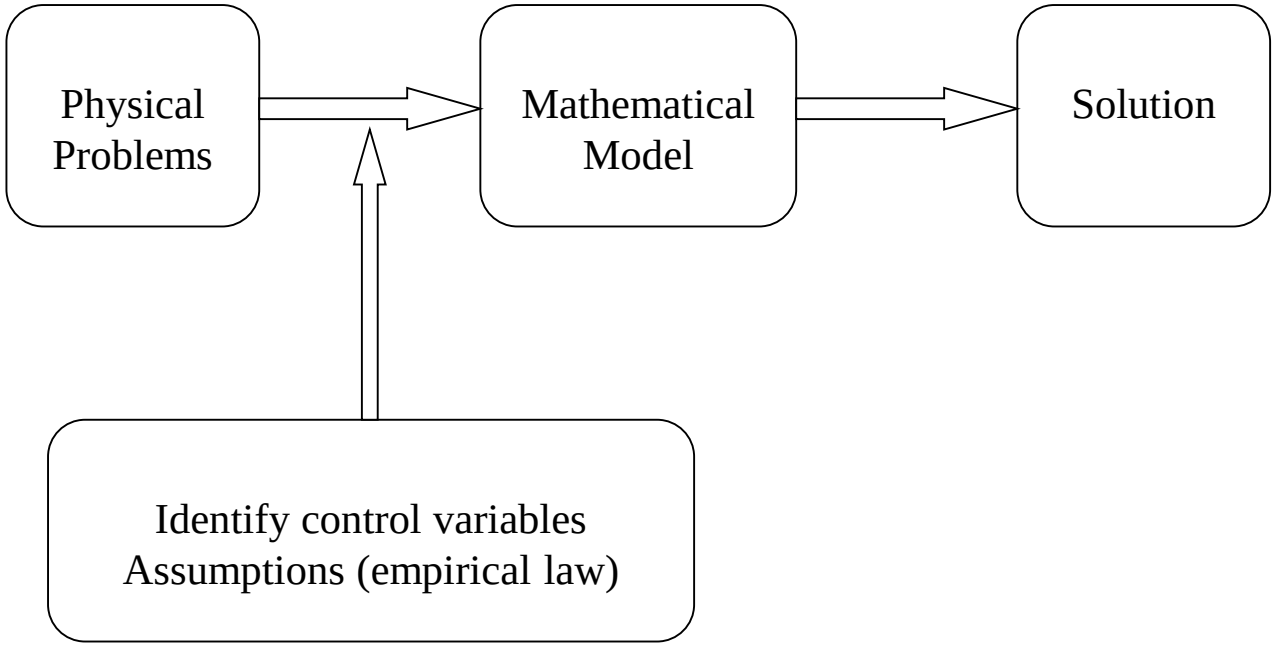
  - Gain familiarity with common algorithms

  - Computing realities and calculations in principle

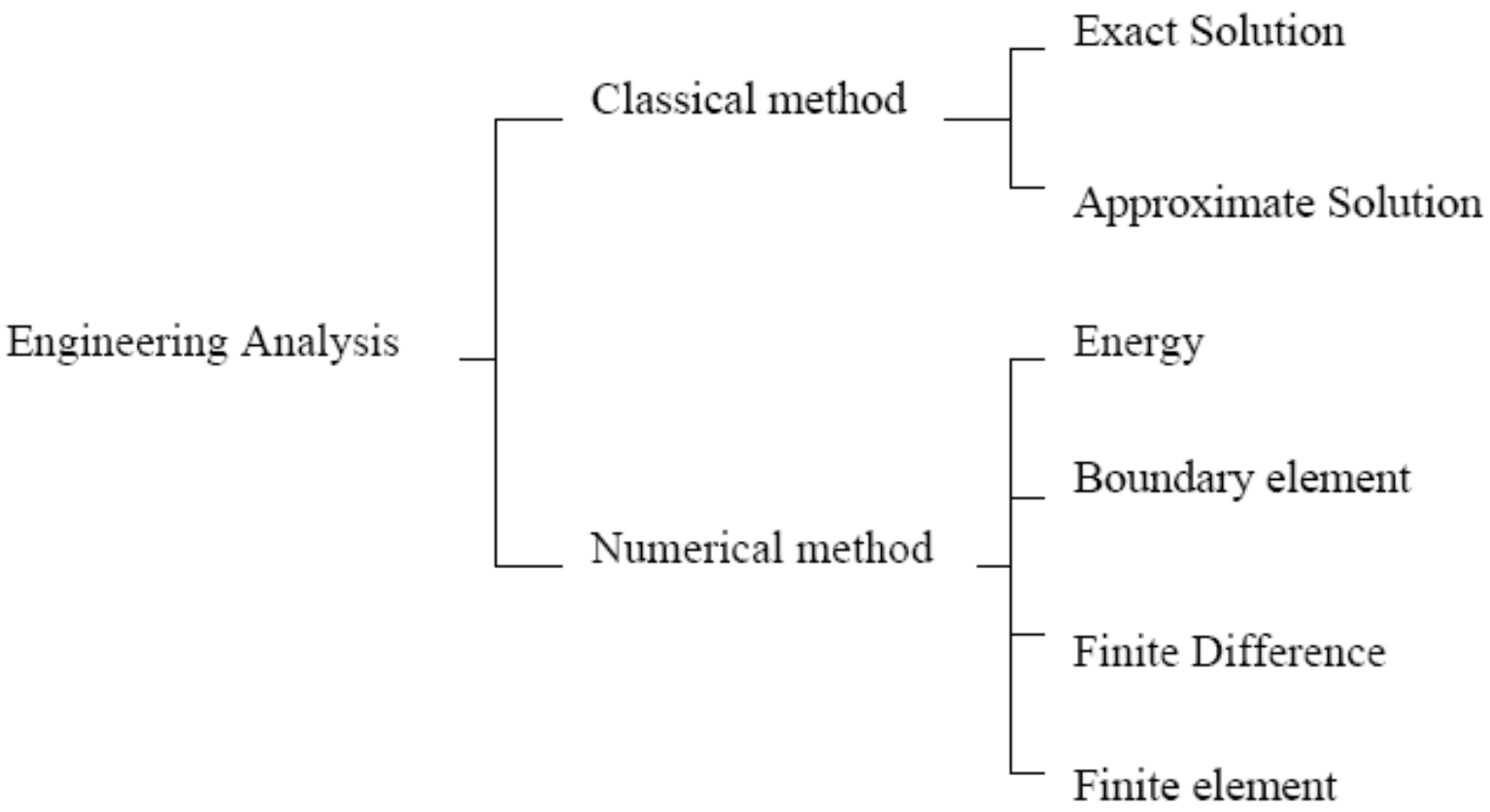
What is Numerical Method

- Numerical method is an approach for solving complex mathematical problem using only simple arithmetic operations
- The approach involves formulation of model of physical situations that can be solved with arithmetic operations
- Step involved
  1. Formulation of mathematical model
  2. Construction of an appropriate numerical method
  3. Implementation of the method to obtain a solution
  4. Validation of the solution

Mathematical model



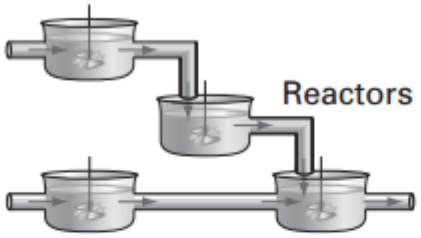
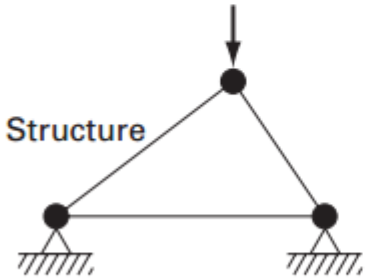
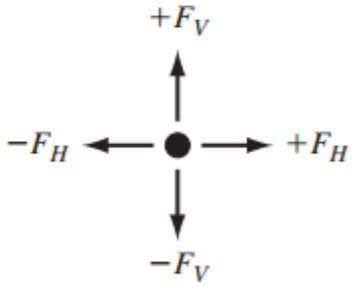
Analysis Method



Characteristics of Numerical Methods

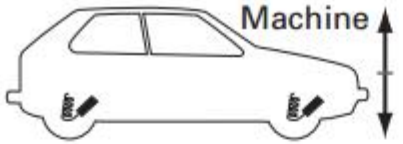
1. The solution procedure is iterative, with the accuracy of the solution improving with each iteration
2. The solution procedure provides only an approximation to the true, but unknown, solution
3. The algorithm is simple and can be easily programmed
4. The solution procedure may occasionally diverge from rather than converge to the true solution

Some applications

Field	Device	Organizing Principle	Mathematical Expression
Chemical engineering		Conservation of mass	Mass balance:  Over a unit of time period $\Delta\text{mass} = \text{inputs} - \text{outputs}$
Civil engineering		Conservation of momentum	Force balance:  At each node Σ horizontal forces $(F_H) = 0$ Σ vertical forces $(F_V) = 0$

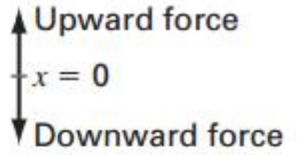
Some applications

Mechanical engineering



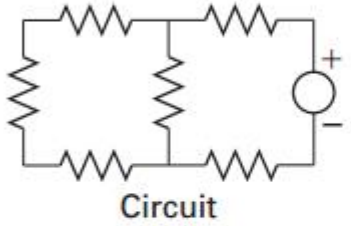
Conservation of momentum

Force balance:



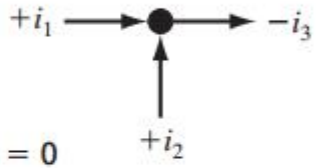
$$m\frac{d^2x}{dt^2} = \text{downward force} - \text{upward force}$$

Electrical engineering



Conservation of charge

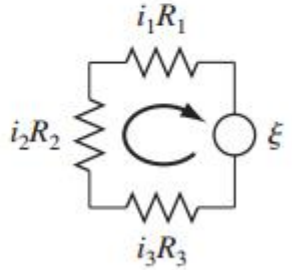
Current balance:



For each node
Σ current $(i) = 0$

Conservation of energy

Voltage balance:



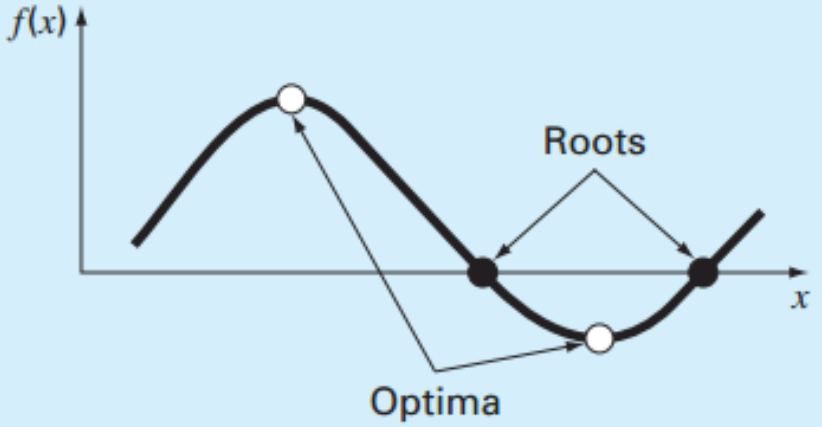
Around each loop
Σ emf's $-$ Σ voltage drops for resistors
$= 0$
$\Sigma\,\xi - \Sigma\,iR = 0$

Some applications

(a) *Part 2*: Roots and optimization

Roots: Solve for x so that $f(x) = 0$

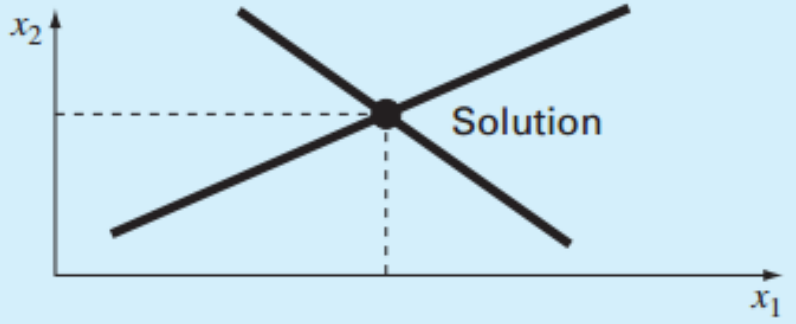
Optimization: Solve for x so that $f'(x) = 0$



(b) *Part 3*: Linear algebraic equations

Given the a's and the b's, solve for the x's

$$a_{11}x_1 + a_{12}x_2 = b_1$$
$$a_{21}x_1 + a_{22}x_2 = b_2$$

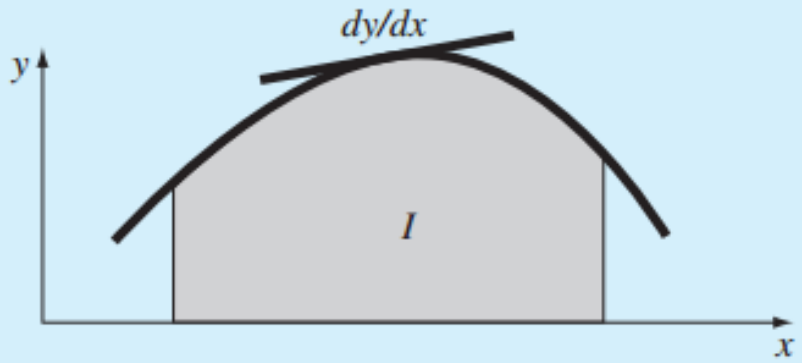


Some applications

(d) *Part 5*: Integration and differentiation

Integration: Find the area under the curve

Differentiation: Find the slope of the curve



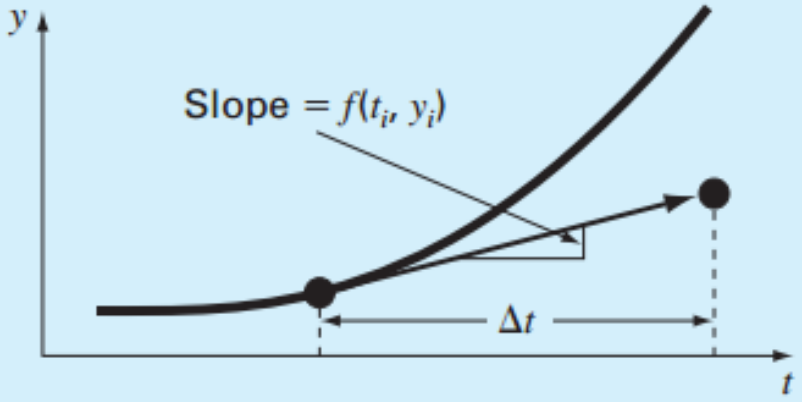
(e) *Part 6*: Differential equations

Given

$$\frac{dy}{dt} \approx \frac{\Delta y}{\Delta t} = f(t, y)$$

solve for y as a function of t

$$y_{i+1} = y_i + f(t_i, y_i)\Delta t$$



Example

Problem Statement. A bungee jumper with a mass of 68.1 kg leaps from a stationary hot air balloon. Use Eq. (1.9) to compute velocity for the first 12 s of free fall. Also determine the terminal velocity that will be attained for an infinitely long cord (or alternatively, the jumpmaster is having a particularly bad day!). Use a drag coefficient of 0.25 kg/m.

$$v(t) = \sqrt{\frac{gm}{c_d}} \tanh\left(\sqrt{\frac{gc_d}{m}} t\right)$$

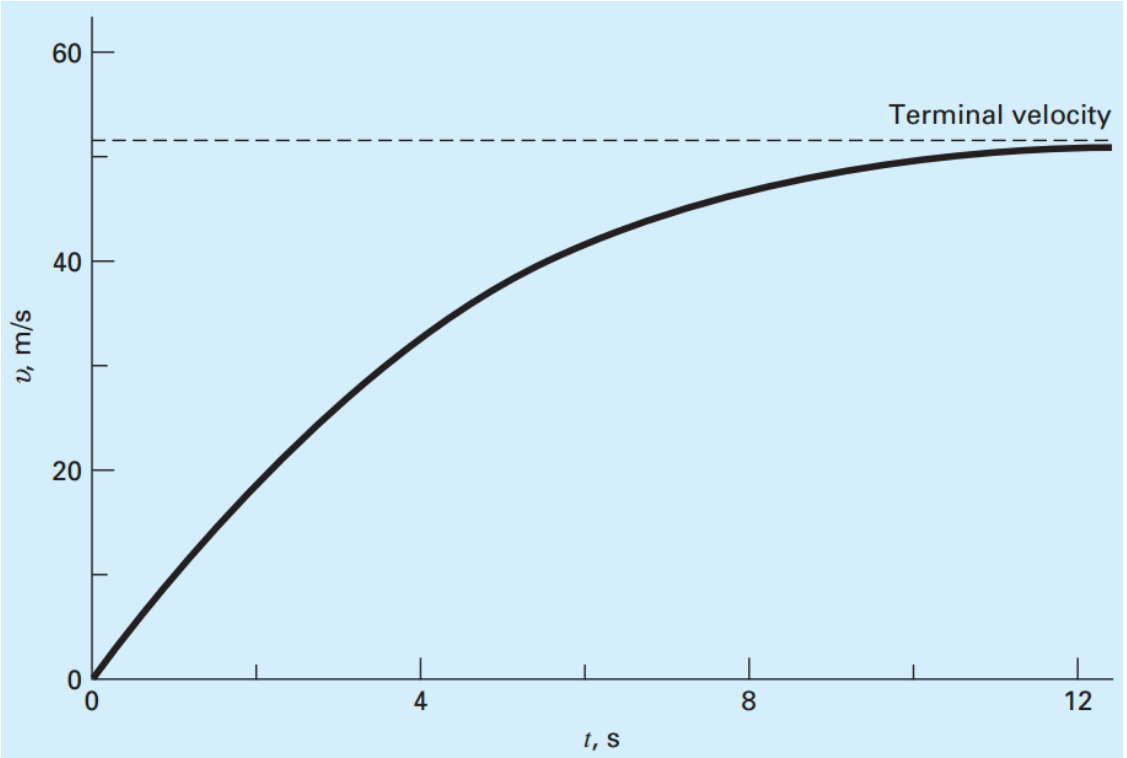
$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$$

Analytical method

$$v(t) = \sqrt{\frac{9.81(68.1)}{0.25}} \tanh\left(\sqrt{\frac{9.81(0.25)}{68.1}}t\right) = 51.6938 \tanh(0.18977t)$$

which can be used to compute

t, s	v, m/s
0	0
2	18.7292
4	33.1118
6	42.0762
8	46.9575
10	49.4214
12	50.6175
∞	51.6938

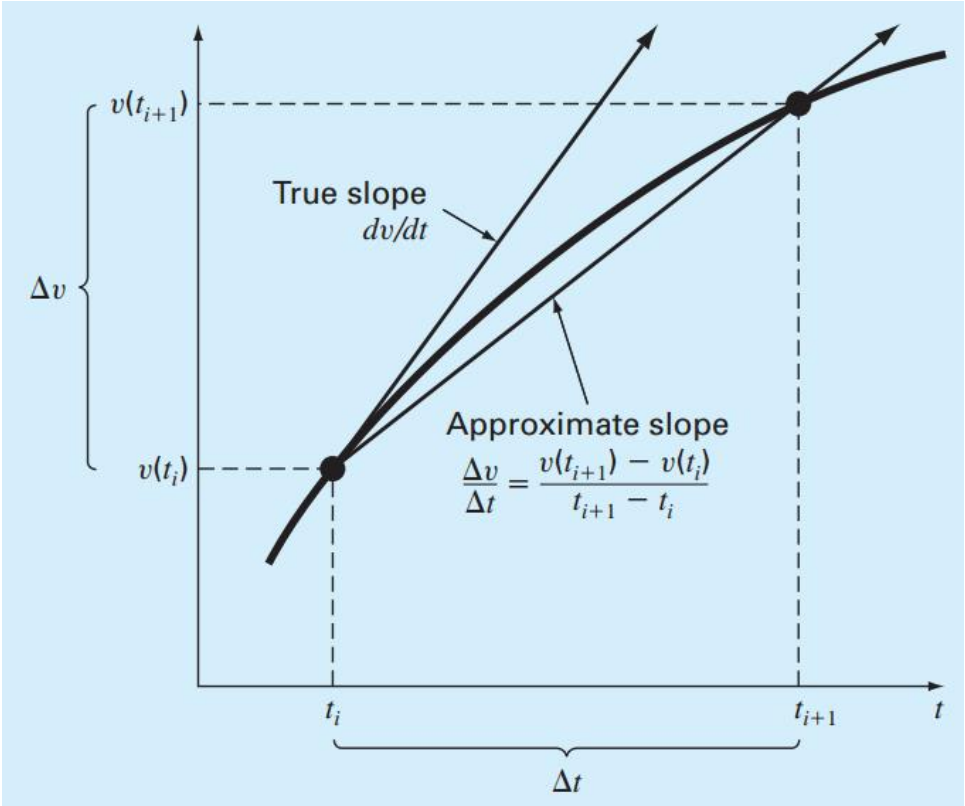


Numerical method

$$v(t) = \sqrt{\frac{gm}{c_d}} \tanh\left(\sqrt{\frac{gc_d}{m}}t\right)$$

$$\frac{dv}{dt} \cong \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}$$

$$\frac{dv}{dt} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t}$$



Numerical method

$$v(t) = \sqrt{\frac{gm}{c_d}} \tanh\left(\sqrt{\frac{gc_d}{m}} t\right)$$

$$\frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i} = g - \frac{c_d}{m} v(t_i)^2$$

This equation can then be rearranged to yield

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c_d}{m} v(t_i)^2\right](t_{i+1} - t_i)$$

$$v_{i+1} = v_i + \frac{dv_i}{dt} \Delta t$$

This approach is formally called *Euler's method*

Numerical method

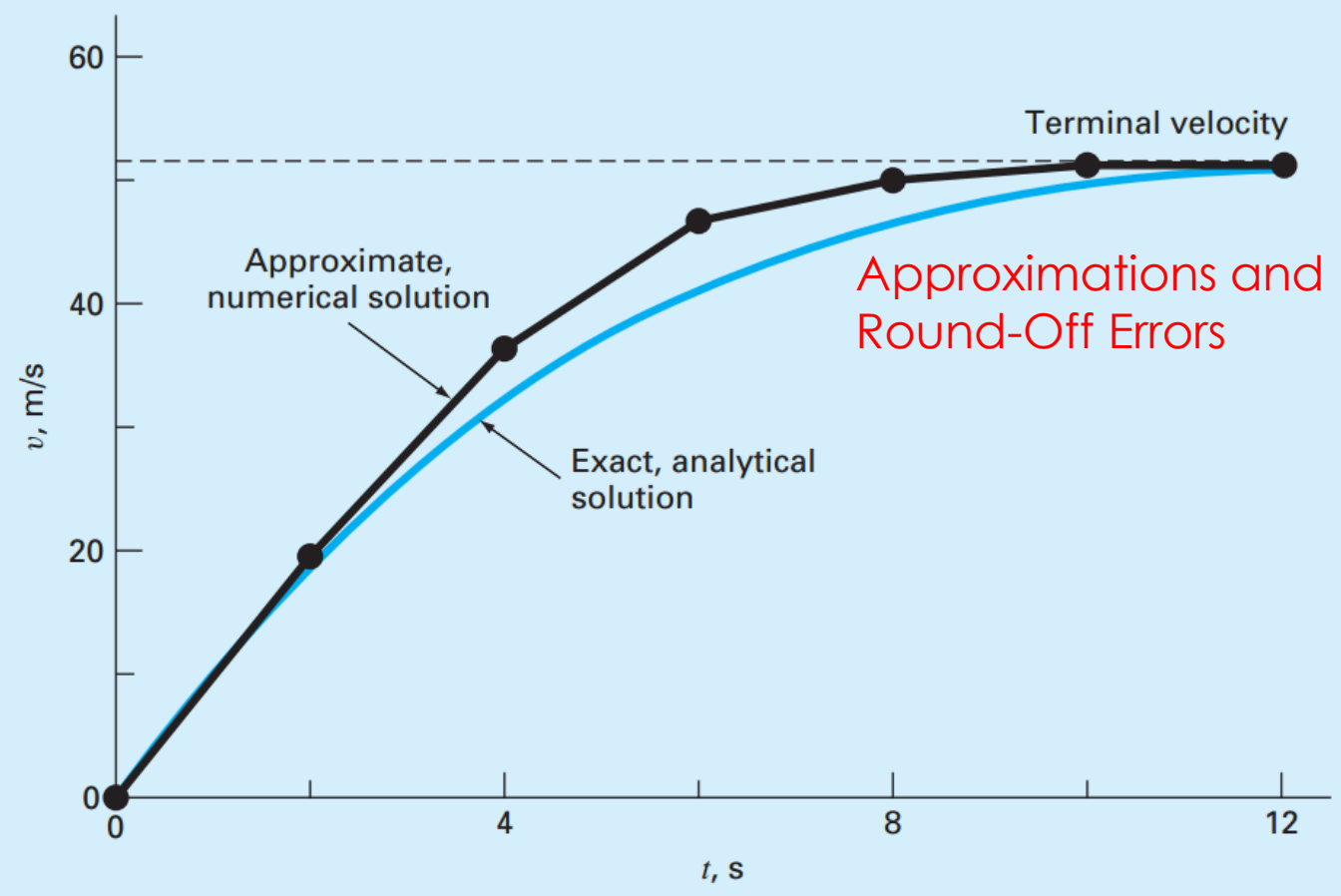
$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c_d}{m} v(t_i)^2\right](t_{i+1} - t_i)$$

at $t_1 = 2$ s $\qquad v = 0 + \left[9.81 - \frac{0.25}{68.1}(0)^2\right] \times 2 = 19.62$ m/s

For the next interval (from $t = 2$ to 4 s)

$$v = 19.62 + \left[9.81 - \frac{0.25}{68.1}(19.62)^2\right] \times 2 = 36.4137 \text{ m/s}$$

Numerical method



Comparison of the numerical and analytical solutions for the bungee jumper problem

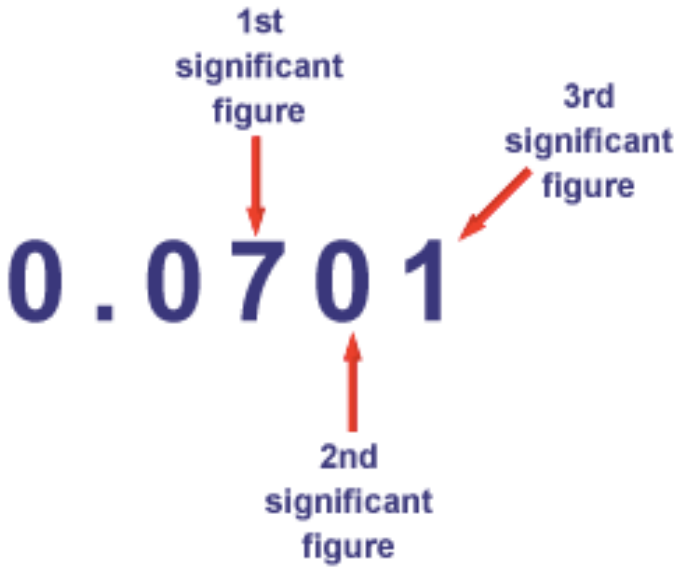
Approximations and Round-Off Errors

1. the numerical method involved an approximation => error
2. the availability of an analytical solution allowed us to compute the error exactly
3. Many applied engineering problems, we cannot obtain analytical solutions => estimates of the errors

Significant figures

- Significant figures (significant digits) of a number are digits in the number that are reliable and absolutely necessary to indicate the quantity of something

- Ex: 0.00001845, 0.0001845, and 0.001845 all have four significant figures.
- Computers retain only a finite number of significant figures => round-off error. Ex: 12.34525 kg => 12.345 kg



ACCURACY AND PRECISION

1. Precision refers to how closely individual computed or measured values agree with each other.
2. Accuracy refers to how closely a computed or measured value agrees with the true value.

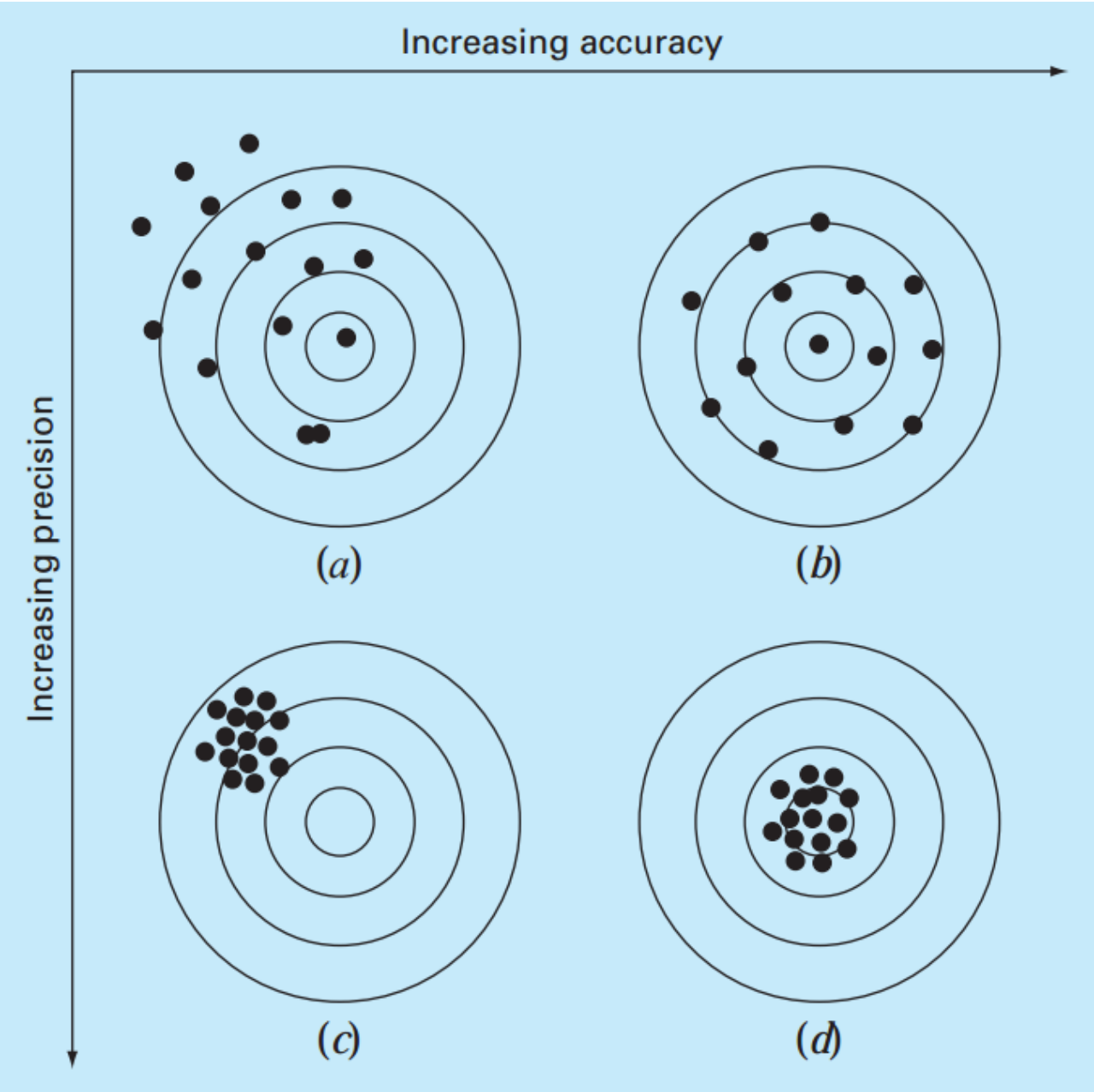
ACCURACY AND PRECISION

(a) Inaccurate and imprecise;

(b) accurate and imprecise;

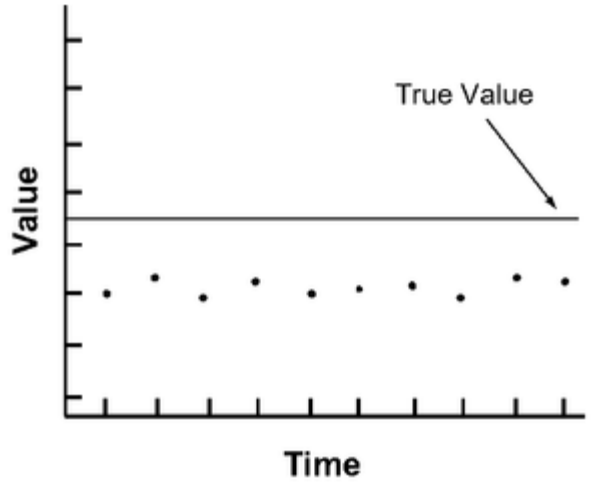
(c) inaccurate and precise;

(d) accurate and precise.



ERROR DEFINITIONS

1. Truncation errors, which result when approximations are used to represent exact mathematical procedures
2. Round-off errors, which result when numbers having limited significant figures are used to represent exact numbers



ERROR DEFINITIONS

- Relationship between the exact, or true, result and the approximation

$$\text{True value} = \text{approximation} + \text{error}$$

$$E_t = \text{true value} - \text{approximation}$$ the "true" error

- The relative error

$$\text{True fractional relative error} = \frac{\text{true error}}{\text{true value}}$$

$$\varepsilon_t = \frac{\text{true error}}{\text{true value}} 100\%$$

⟺ with functions that can be solved analytically.

ERROR DEFINITIONS

with functions that can not be solved analytically.

$$\varepsilon_a = \frac{\text{approximate error}}{\text{approximation}} 100\%$$

the "true" error

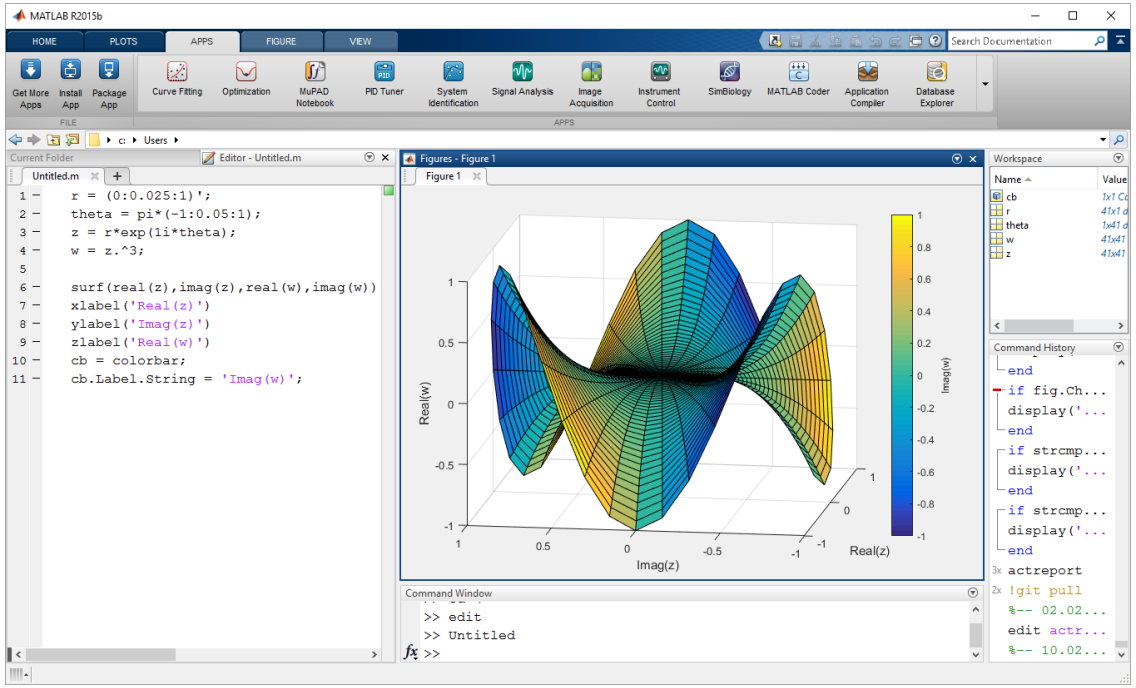
$$\varepsilon_a = \frac{\text{current approximation} - \text{previous approximation}}{\text{current approximation}} 100\%$$

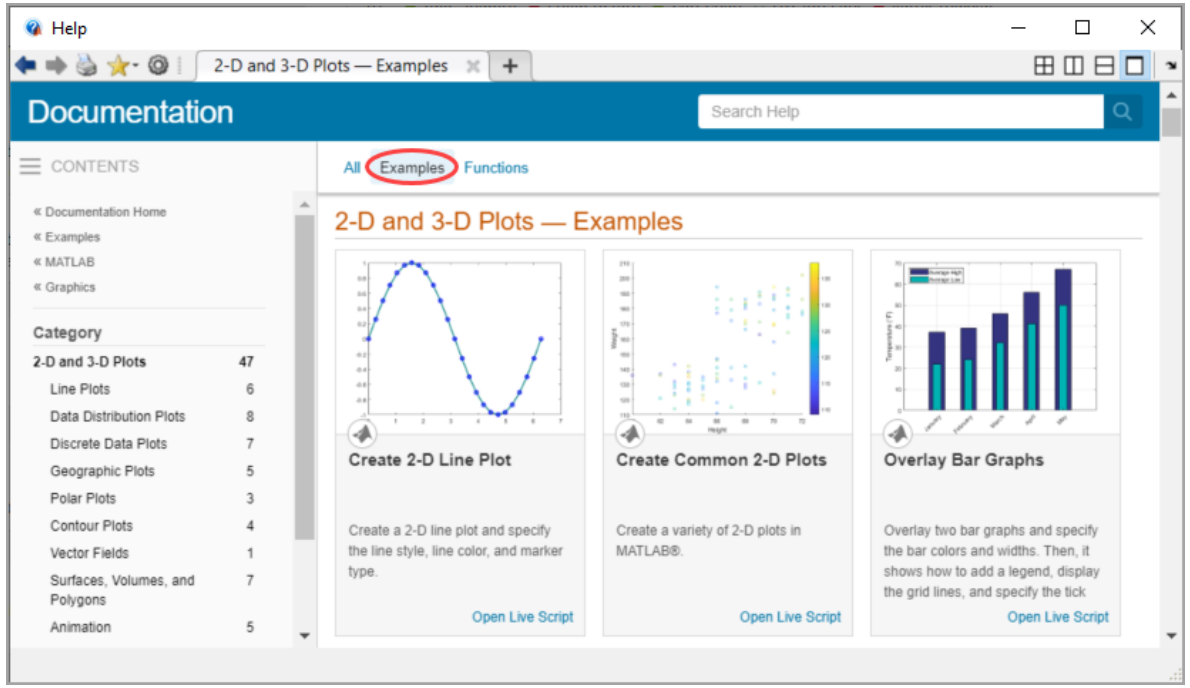
a signifies that the error is normalized to an approximate value

For such cases, the computation is repeated until

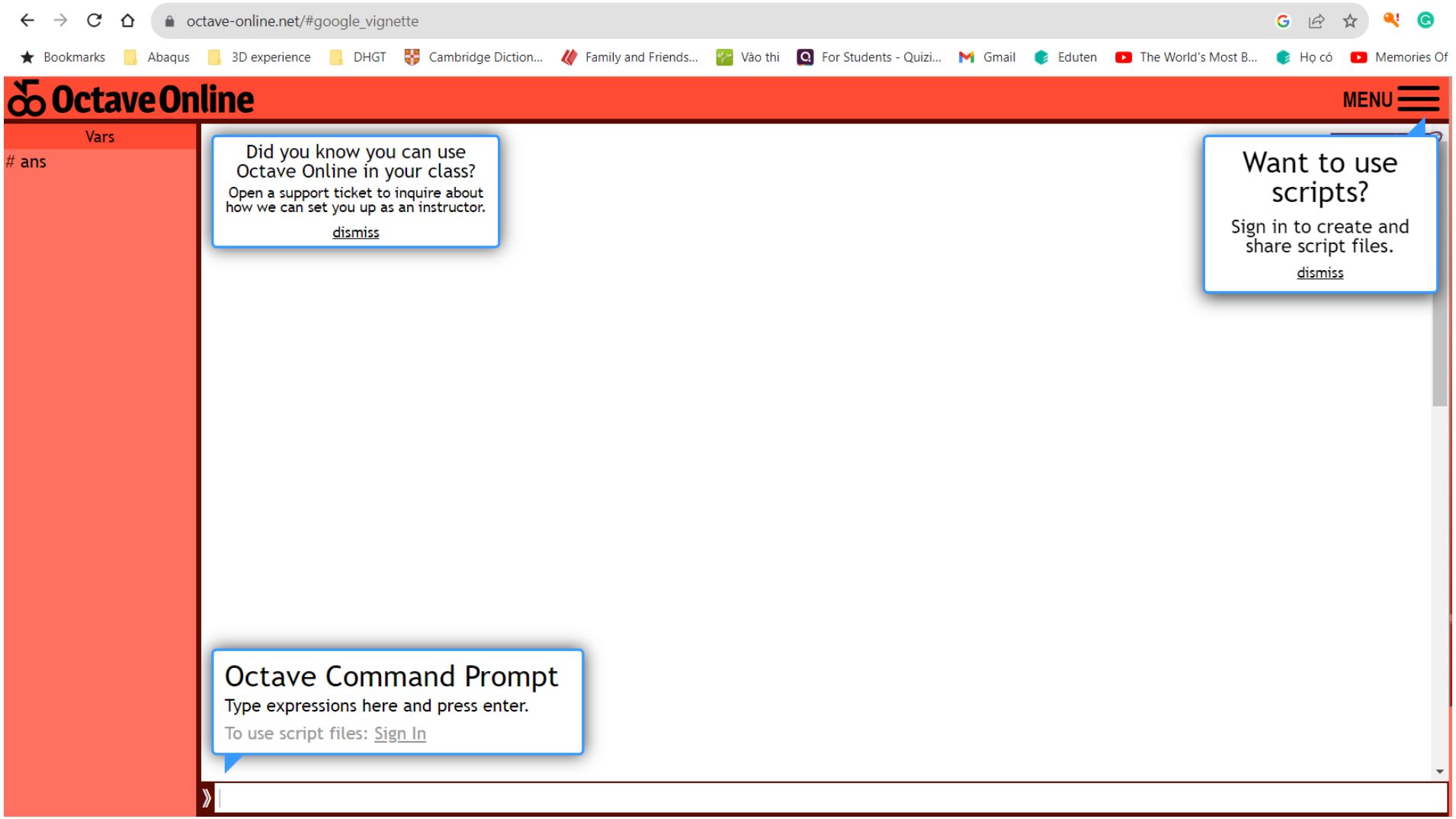
$$|\varepsilon_a| < \varepsilon_s$$ prespecified percent tolerance

Matlab/Octave introduction





Matlab/Octave introduction



Matlab introduction



- Stands for MATrix LABoratory
- Interpreted language
- Scientific programming environment
- Very good tool for the manipulation of matrices
- Great visualisation capabilities
- Loads of built-in functions
- Easy to learn and simple to use

Matlab introduction

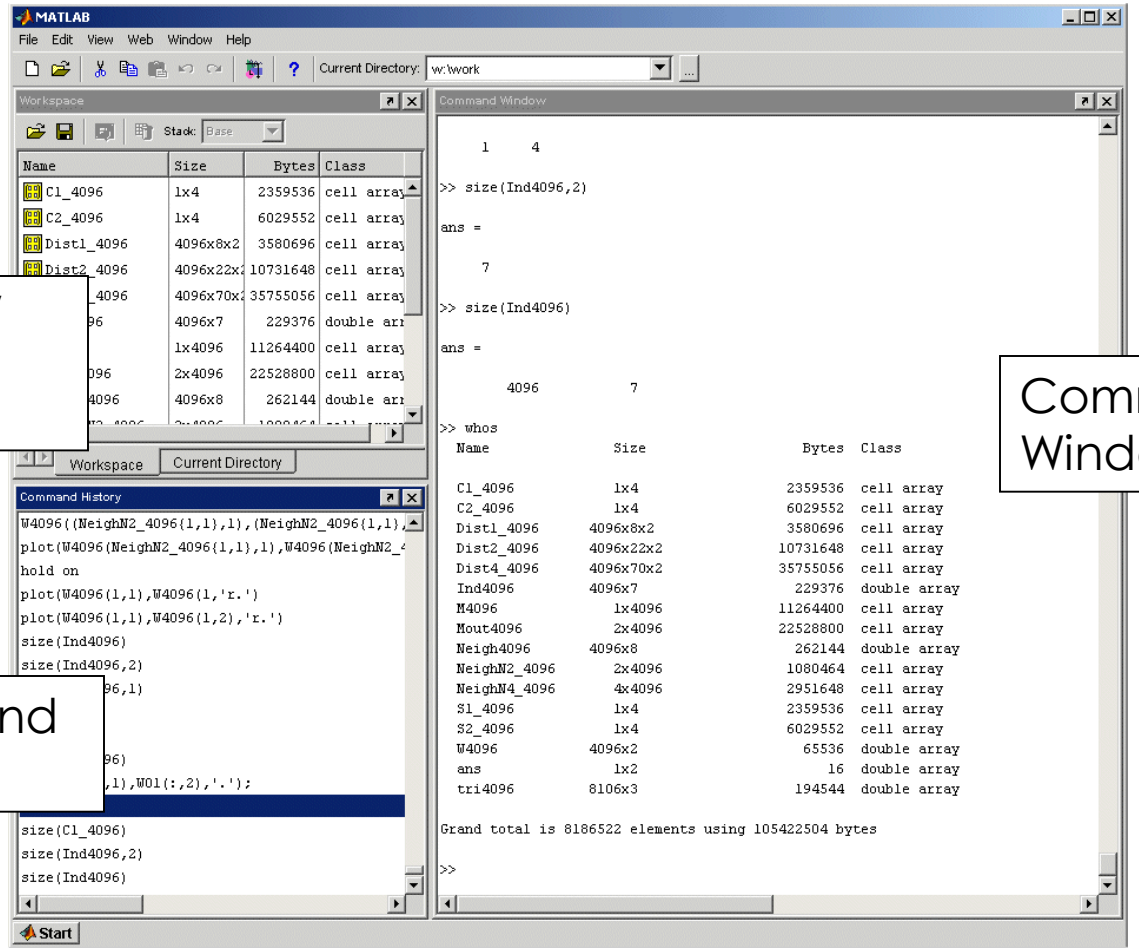
MATLAB software

- A convenient environment for performing many types of calculations
- A very nice tool to implement numerical methods

MATLAB uses three primary windows:

- Command window. Used to enter commands and data.
- Graphics window. Used to display plots and graphs.
- Edit window. Used to create and edit M-files.

Matlab Desktop



Workspace / Current Directory

Command History

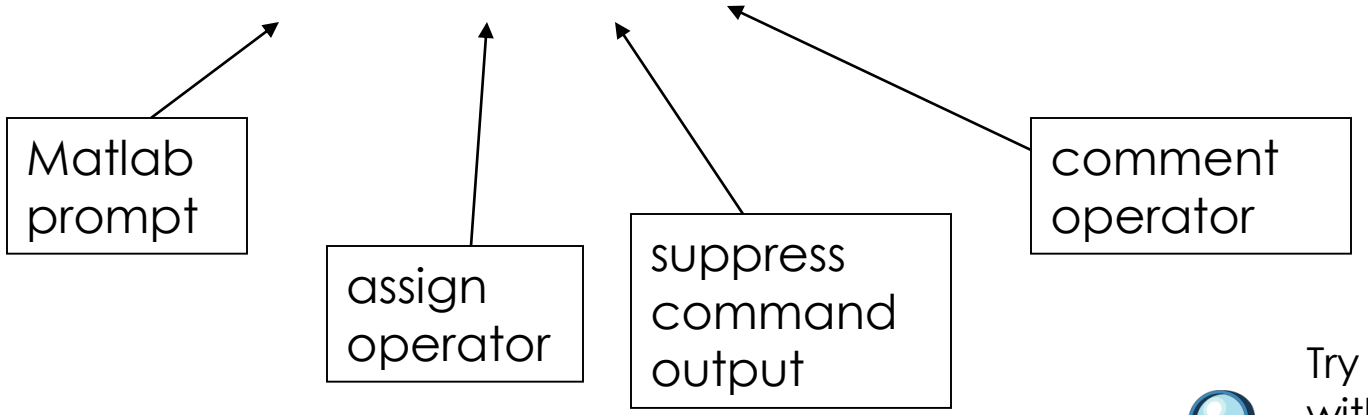
Command Window

Explore the Matlab Desktop

Variables

- Don't have to declare type
- Don't even have to initialise
- Just assign in command window

```
>>
>> a=12; % variable a is assigned 12
```





Try the same line without the semicolon and comments

Variables (continued …)

- View variable contents by simply typing the variable name at the command prompt

```
>> a
a =
        12
>>
>> a*2
a =
        24
>>
```

Matlab introduction

Mathematical operations

- Type 2+3 after the >> prompt, followed by **Enter** (press the **Enter** key) as indicated by <**Enter**>:
  >> 2+3 < **Enter**>
- Commands are only carried out when you enter them. The answer in this case is, of course, 5. Next try
  >> 3–2 < **Enter**>
  >> 2*3 < **Enter**>
  >> 1/2 < **Enter**>
  >> 2^3 < **Enter**>
  >> 2\1 < **Enter**>

^	Exponentiation
–	Negation
* /	Multiplication and division
\	Left division[2]
+ –	Addition and subtraction

Workspace

- The workspace is Matlab's memory
- Can manipulate variables stored in the workspace

```
>> b=10;
>> c=a+b
c =
    22
>>
```

Workspace (continued …)

- Display contents of workspace

```
>> whos
 Name      Size                Bytes  Class
  a        1x1                     8  double array
  b        1x1                     8  double array
  c        1x1                     8  double array
Grand total is 3 elements using 24 bytes
>>
```

- **Delete variable(s) from workspace**

```
>> clc
>> clear all; % delete a and b from workspace
>> close all; % delete all variables from workspace
```

Matlab help commands

- help

>> help sum % displays documentation for the function sum

>> lookfor convert % displays functions with convert in the first help line

- Start Matlab help documentation

>> helpdesk

Matlab introduction

- **Vector**
  - ✓ a = [1 2 3 4 5] or >> a = 1:5
    and
  - ✓ a = [1 ; 2 ; 3 ; 4 ; 5] or a = [1 2 3 4 5]'

- **Matrix**
  - ✓ A = [1 2 3; 4 5 6; 7 8 9]
  - ✓ z = zeros(2,4)
  - ✓ x = ones(2,3)

$$z = zeros(2, 4)$$

$$z = \begin{matrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{matrix}$$

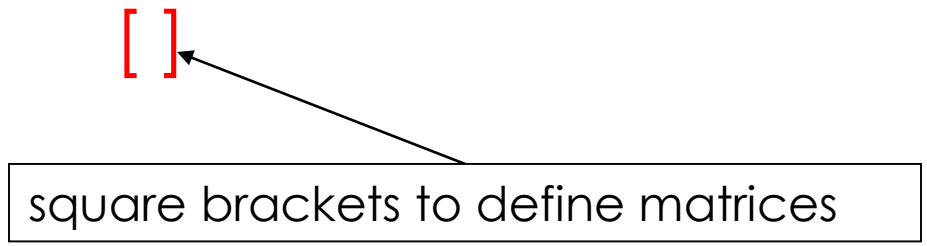
$$x = ones(2, 3)$$

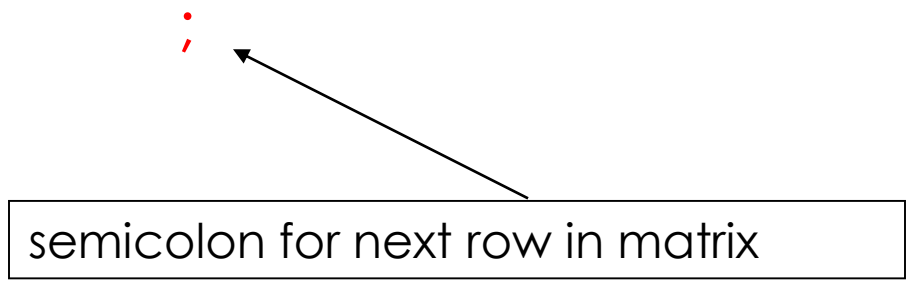
$$x = \begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{matrix}$$

Matrices

- Don't need to initialise type, or dimensions

```
A = [3 2 1; 5 1 0; 2 1 7]
A =
     3     2     1
     5     1     0
     2     1     7
```





Manipulating Matrices

A =

3	2	1
5	1	0
2	1	7

- Access elements of a matrix

A(1,2)

ans=

2



- Remember Matrix(row,column)
- Naming convention Matrix variables start with a capital letter while vectors or scalar variables start with a simple letter

The : operator

- VERY important operator in Matlab
- Means ‘to’

```
>> 1:10
ans =
     1     2     3     4     5     6     7     8     9    10
>> 1:2:10
ans =
     1     3     5     7     9
```

🔎 Try the following

```
>> x=0:pi/12:2*pi;
>> y=sin(x)
```

The : operator and matrices

```
A(3,2:3)
ans =
     1     7
A(:,2)
ans =
     2
     1
     1
```

```
A =
     3     2     1
     5     1     0
     2     1     7
```



What'll happen if you type A(:,:) ?

Manipulating Matrices

```
A =
     3     2     1
     5     1     0
     2     1     7
```

```
B =
     1     3     1
     4     9     5
     2     7     2
```

```
>> A '      % transpose
>> B*A      % matrix multiplication
>> B.*A     % element by element multiplication
>> B/A      % matrix division
>> B./A     % element by element division
>> [B A]    % Join matrices (horizontally)
>> [B; A]   % Join matrices (vertically)
```



Enter matrix B into the Matlab workspace

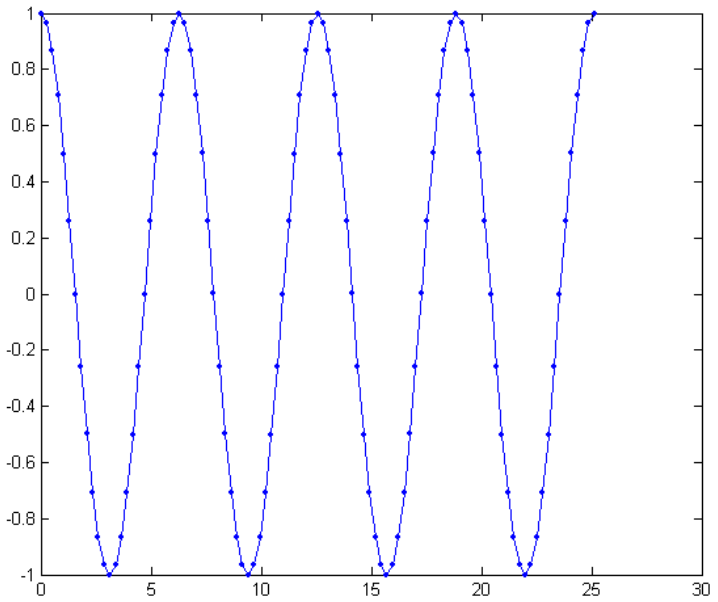


Create matrices A and B and try out the the matrix operators in this slide

Visualisation - plotting data

```
>> figure   % create new figure
>> t=0:pi/12:8*pi;
>> y=cos(t);
>> plot(t,y,'b.-')
```

Plot style



Investigate the function
>> y=A*cos(w*t+phi);
for different values of phi (eg: 0, pi/4, pi/3, pi/2), w (eg: 1, 2, 3, 4) and A (eg: 1, 0.5, 2). Use the **hold on** Matlab command to display your plots in the same figure. Remember to type **hold off** to go back to normal plotting mode. Try using different plot styles (**help plot**)

A = amplitude
phi = phase
w = angular frequency = 2*pi*frequency

```
2
3 -    pop_size = 1e4; % Population size
4
5      % Generate population distribution through code (can also be done using randtool GUI):
6      % pop_in = rand(pop_size,1); % Uniform distribution over interval (0,1)
7      % pop_in = 100*rand(pop_size,1); % Uniform popuation over interval (0,100)
8 -    pop_in = random('Rayleigh',2,pop_size); % Rayleigh distribution with size parameter 2
9      % pop_in = randn(pop_size,1); % Standard normal distribution
10
11 -   N1 = 5; % Sample size 1
12 -   N2 = 15; % Sample size 2
13 -   N3 = 30; % Sample size 3
14 -   N4 = 60; % Sample size 4
15
16 -   M = 200; % How many times to draw random samples from the population
17
18 -   means_N1 = zeros(M,1); % For N1
19 -   for i = 1:M % Loop over repeat drawing of samples
20 -       tempdraw = pop_in(randperm(pop_size, N1));
21 -       means_N1(i) = mean(tempdraw);
22 -   end
23
24 -   means_N2 = zeros(M,1); % For N2
25 -   for i = 1:M % Loop over repeat drawing of samples
26 -       tempdraw = pop_in(randperm(pop_size, N2));
27 -       means_N2(i) = mean(tempdraw);
28 -   end
29
30 -   means_N3 = zeros(M,1); % For N3
31 -   for i = 1:M % Loop over repeat drawing of samples
32 -       tempdraw = pop_in(randperm(pop_size, N3));
33 -       means_N3(i) = mean(tempdraw);
34 -   end
35
36 -   means_N4 = zeros(M,1); % For N4
37 -   for i = 1:M % Loop over repeat drawing of samples
38 -       tempdraw = pop_in(randperm(pop_size, N4));
39 -       means_N4(i) = mean(tempdraw);
40 -   end
41
42 -   figure, hist(pop_in,30), title('Population Distribution')
43 -   figure
44 -   subplot(2,2,1), hist(means_N1,30), title('Distribution of the mean - N = 5')
45 -   subplot(2,2,2), hist(means_N2,30), title('Distribution of the mean - N = 15')
46 -   subplot(2,2,3), hist(means_N3,30), title('Distribution of the mean - N = 30')
47 -   subplot(2,2,4), hist(means_N4,30), title('Distribution of the mean - N = 60')
```





Debugging

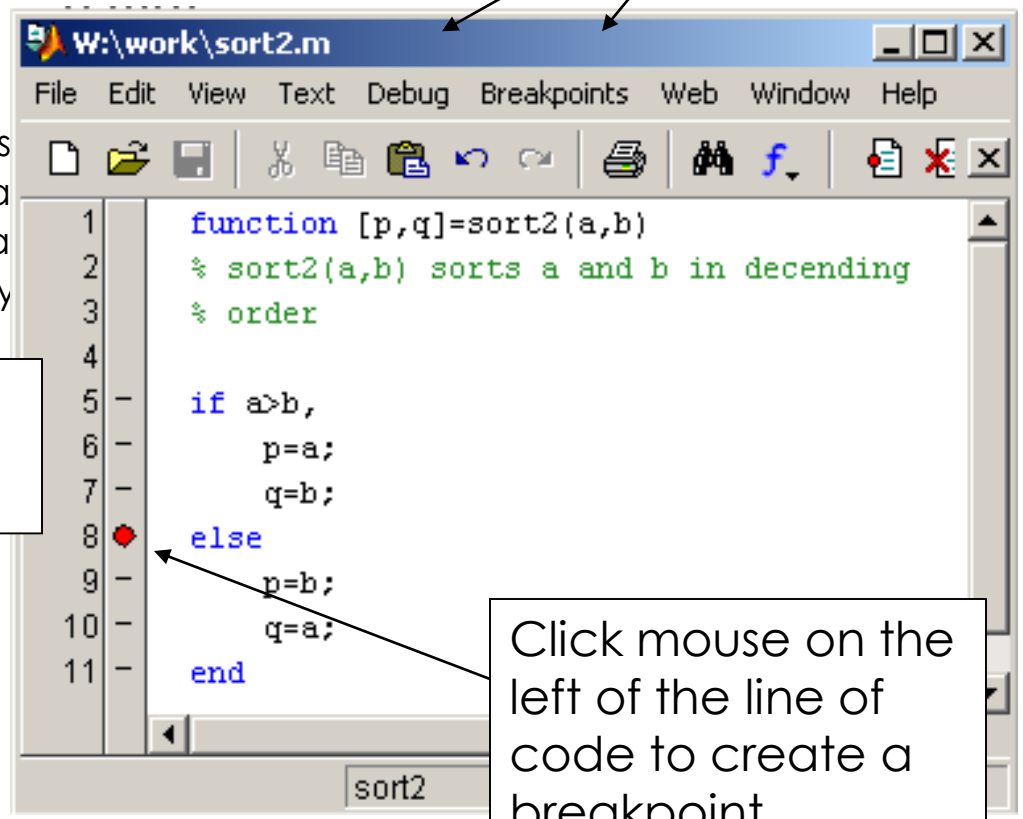
- Set breakpoints to stop the execution of code

```
>> [i j]=sort2(2,4)
K>>
K>> whos
 Name      Size                Bytes  Class
  a         1x1                     8  double a
  b         1x1                     8  double a
Grand total is 2 elements using 16 by
K>> a
a =
     2
K>> return
i =

     4
j =
     2
```

local function workspace

exit debug mode



Matlab introduction

- Some commonly used tools and commands:
  - ↑ (up arrow) returns last command input, can be repeated
  - clc – clears the screen
  - clear all – clears variables
  - close all – closes figures

Octave introduction

Octave online:

- https://rextester.com/l/octave_online_compiler
- https://www.tutorialspoint.com/execute_matlab_online.php

Exercise 6: Create a matrix A that:

- has size of 3*3
- all the entries is 1

a, Create matrix b that is transpose of **A**
b, Compute sum of **A** and **B**
c, Compute **A*B** and **B*A**
d, Increment all the elements of A by 2.